TerraTrace: Reproducible, Uncertainty-Aware, Multi-Lens Morphological Analog Screening Across Five Worlds

NASA Space Apps 2026 - research preview. Repository: `github.com/pal-123456789/terratrace`

> **Scope / claim firewall.** TerraTrace screens terrain *morphology* between curated map products. It does **not** estimate rover traversability, validate autonomy, establish geological equivalence, certify site access, or certify mission readiness. Earth sites named here are *illustrative analogs*, not endorsed field sites. It is not NASA-endorsed.

Abstract

Choosing a terrestrial analog for a planetary terrain is usually done by expert judgment and reported as a single qualitative verdict. TerraTrace makes that screening **quantitative, reproducible, and honest**. It brings a planetary reference terrain and candidate Earth terrains to a common metric support on each body's own geometry, computes a battery of map-scale morphology descriptors, and ranks candidates by **Pareto dominance across several independent descriptor "lenses"** - deliberately refusing to collapse the comparison into one composite "likeness" score. When lenses disagree it returns an explicit TIE / INDETERMINATE state instead of inventing a winner, and attaches a spatial block-bootstrap, a permutation significance test, a non-composite decision-confidence verdict, and a full provenance record to every result. On a real five-world run (Mars MOLA vs three Copernicus GLO-30 Earth terrains) the method independently re-identifies a documented basalt analog - Craters of the Moon, Idaho - as the closest match on the core roughness/relief axes, while honestly reporting that the recommendation does not survive every lens. Everything is offline-reproducible from one command and covered by 771 automated tests.

1. Problem and approach

Analog site selection underpins instrument testing, operational rehearsal, and the interpretation of orbital data. Yet "X is a good analog for Y" is rarely stated with a defined support scale, an uncertainty, or a falsifiable procedure. TerraTrace treats analog screening as a transparent, auditable computation with three commitments: (i) **comparison at a declared common support** on each body's correct geometry; (ii) **no composite score** - candidates are ranked by Pareto dominance over interpretable axes, and ties are reported, not broken; and (iii) **every number carries provenance and uncertainty**. The goal is not to declare a winner but to say, defensibly, *what transfers, at what scale, and with what confidence.*

2. Method

2.1 Common support and per-body reprojection. Each area of interest is warped to its own local azimuthal-equidistant metric grid using the correct body radius (Mars 3,396,190 m; Moon 1,737,400 m; Mercury 2,439,400 m; Ceres 469,730 m; Earth WGS84) and aggregated to a declared common support (e.g. 1000 m), so descriptors compare like with like rather than conflating pixel size with terrain.

2.2 Morphology descriptors. Core: slope, local relief, detrended ruggedness, low-slope connectivity. Extended: aspect dispersion, hypsometric integral. Geomorphic: Zevenbergen-Thorne curvature and the Vector Ruggedness Measure. Landform: geomorphon landform composition and topographic openness. Scale & fabric: the radially-averaged power-spectral slope and a directional-structure-function anisotropy (fabric strength and orientation). Planetary roughness: the Kreslavsky-Head (2000) differential-slope baseline roughness (the metric behind NASA's global MOLA roughness maps) and the Iwahashi-Pike (2007) three-signature (slope / convexity / texture) nested-means terrain class. Dissection: a depression-filled D8 flow-accumulation surface-dissection texture (a purely geometric routing descriptor). The bank also carries a Haar wavelet multiscale-energy spectrum, the Strahler hypsometric curve, surface-normal-dispersion roughness, and a Weiss (2001) TPI slope-position classification. A Topographic Wetness Index (Beven & Kirkby 1979) and a convergence index (Kothe & Lehmeier 1996) summarise flow-geometric convergence; a sky-view factor (Zaksek et al. 2011) captures hemispherical openness; and a two-scale TPI slope-position classification (Weiss 2001) resolves nested landform position. Finally a differential box-counting fractal dimension, gliding-box lacunarity, Shannon terrain entropy, and a Flint (1974) slope-vs-contributing-area concavity complete the battery. A geostatistical semivariogram (range/sill/nugget; Matheron 1963) and an RMS/Allan-deviation roughness length (Shepard et al. 2001) summarise spatial structure; Wood (1996)

morphometric curvature classes and a Woodcock (1977) surface-normal orientation-tensor fabric add shape descriptors; an openness-difference slope position (Yokoyama 2002) and the Melton (1965) ruggedness number complete the batch. Distributions are compared to the reference with scale-normalized Wasserstein/KS distances.

2.3 Decision: Pareto, not a score. Candidates become vectors of distances-to-reference and are ranked by **Pareto dominance** (peeled non-dominated layers + a dominance matrix). The result is one of `RECOMMEND_FOLLOW_UP`, `TIE`, `INDETERMINATE`, `NOT_COMPARABLE`, or `BLOCKED`. Six **lenses** - core, extended, geomorphic, landform, texture, hydro - are independent axis sets (the hydro lens adds a Topographic Wetness Index, aspect convergence, and flow-dissection distance-to-reference axes); a recommendation that holds under more lenses is more robust, and lens disagreement is surfaced, not hidden.

2.4 Uncertainty, significance, confidence. A spatial moving-block bootstrap gives autocorrelation-aware distance intervals and a non-domination frequency; a one-sided permutation test asks whether the recommended unit is genuinely closer to the reference than the runner-up; and a **non-composite decision-confidence** verdict (`robust` / `qualified` / `fragile` / …) restates the bootstrap, significance, and lens-stability evidence as three independent signals, explicitly naming any signal it could not evaluate.

2.5 Provenance and reproducibility. Every study is hash-anchored (sha256), carries an evidence graph and a transfer-gap ledger, and exports a self-contained bundle (study package, Markdown + offline-HTML report, viewer, references, checksums, confidence). A fixed synthetic study is byte-reproducible within an environment; portable invariants (not pinned digests) gate it across platforms. The full geomorphometric descriptor bank and a swath/profile `transect` extractor are reachable from the unified CLI (`bank`, `transect`) and the loopback-only HTTP API (`POST /bank`, `POST /transect`, the same host-allowlist and body-cap guards as the existing endpoints), and the terrain fingerprint carries an opt-in geostatistical similarity term (`include_geostat`).

3. Data

Mars reference: MGS **MOLA 463 m** (USGS Astrogeology, keyless). Earth candidates: **Copernicus GLO-30** via OpenTopography. The engine's source catalog carries 13 products across the five bodies with coverage, licence, and 1-sigma vertical-accuracy metadata; a no-network preflight validates each bounding box, selects the covering product, and sizes the request before any download.

4. Result: a real five-world screen

Mars MOLA versus three real Copernicus GLO-30 Earth terrains at 1000 m support. Distance to the Mars reference per axis (lower = closer):

candidate	slope	relief	ruggedness	connectivity	aspect	hypsometric
Craters of the Moon, ID	**0.75**	**0.78**	**0.76**	**0.018**	0.39	0.28
Nevada basin	1.33	1.38	1.45	0.030	**0.011**	**0.14**
Andean volcano	1.73	1.50	1.85	0.057	0.012	0.17

Four-lens ablation on the **same** data (one command, `--ablation`):

lens	result	recommended	non-dominated set
core	RECOMMEND	**Craters of the Moon**	{Craters}
extended	TIE	-	{Craters, Nevada}
geomorphic	RECOMMEND	**Craters of the Moon**	{Craters}
landform	TIE	-	{Craters, Nevada, Andes}

Reading. On raw roughness/relief/ruggedness (core) and on curvature/vector-ruggedness (geomorphic), Craters of the Moon is decisively the closest match - the method *independently re-identifies a terrain already used as a basalt analog.* Adding aspect-dispersion and hypsometric structure (extended), where the Nevada basin matches Mars better and even shares its ~45 deg fabric, turns the decision into an honest TIE; the landform lens cannot separate the three. The defensible conclusion is therefore **lens-conditional**: *Craters of the Moon is the robust*

analog for surface roughness and local shape; it is not uniquely best for directional/elevation structure. No single number could have said that honestly.

Corroboration at scale. Running the one-command atlas over four distinct Mars terrains (Jezero and Gale crater floors; Daedalia Planum and Elysium Planitia volcanic plains) against the same three Earth candidates under all four lenses (16 referencexlens cells) makes the result sharper: Craters of the Moon is Pareto non-dominated in **every** cell - the sole consensus analog - and is the *decisive* recommendation for the volcanic plains under three of four lenses, while the crater floors tie more often (no clean crater-floor analog among the three). The engine points confidently where the morphology agrees and abstains where it does not. See `REAL-EXAMPLE.md`.

5. Validation

Correctness is pinned by a scientific-correctness suite of 51 oracles that drive the real engine on synthetic terrains with known answers: self-distance is zero; slope/relief/ruggedness are invariant to a constant offset and to 90 deg rotation; roughness discrimination is monotone; a planted close twin is recovered over an impostor; a water tile is BLOCKED; openness is 90 deg on a plane and exceeds 90 deg at a peak; geomorphons label a cone apex a peak and a basin centre a pit; a sinusoid's spectral peak lands at its wavelength; white noise is spectrally flat (Parseval holds); a ridged surface reports the correct fabric orientation. The real-data run adds an external check: the method re-identifies a documented basalt analog on the core axes without being told to.

6. Limitations and transfer gaps (stated, not hidden)

TerraTrace compares *map-scale morphology of sampled DEMs* and nothing else. It says nothing about composition, rock-scale obstacles below the pixel, trafficability, or mission suitability; those bounds are reproduced verbatim in every report. Two concrete limits of the worked example: the Mars window (~10 deg) and the Earth windows (~0.4 deg) differ in footprint by ~15x, so the screen mixes regional Mars structure with local Earth structure (a transfer gap the ledger flags; the fix is comparable-footprint AOIs); and the independent-pixel bootstrap is optimistic relative to the spatial block bootstrap on autocorrelated terrain. Deferred work: DEM co-registration and vertical-datum error propagation end-to-end into the decision, and a multi-pair "analog atlas."

7. Reproducibility

```
pip install -r terratrace/requirements.txt      # core: NumPy only
python terratrace/tests/run_tests.py             # 771 offline tests
python terratrace/scripts/run_validation.py      # 51 correctness oracles
# real five-world study, one command (needs a free OpenTopography key):
python terratrace/scripts/run_real_analog_study.py --config config/real_multibody.example.yml
```

CI runs the suite and the validation oracles on Python 3.10/3.11/3.12. Large DEMs and generated bundles are git-ignored; everything else regenerates deterministically.

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Morphology screening and resampling statistics only - not a traversability, geologic-equivalence, site-access, or mission-readiness claim.